

## Validation against published sources



*mortgagemath 0.6.1 · rendered 2026-05-06*

Every fixture in the table below reproduces **every** value its source publishes, to the cent. The library follows a strict policy: if any published value diverges, the fixture stays out. The six parameter columns show the exact `LoanParams` settings required to match each source — the same six knobs documented in the *at-a-glance* vignette.

### Validated fixtures

| Source                   | Loan                              | Day    | Bal | Rnd | Cmp            | Freq | ARM    |
|--------------------------|-----------------------------------|--------|-----|-----|----------------|------|--------|
| Arcones SOA FM §4.1 Ex 4 | \$20k / 8% / 12yr                 | 30/360 | RE  | HU  | A              | A    | —      |
| BCC Brescia IT           | \$200k / 5.65% / 10yr             | 30/360 | RE  | HU  | M              | M    | —      |
| Broverman MIC 2.7(a)     | \$12k / 12% / 3yr                 | 30/360 | RE  | HU  | M              | M    | —      |
| Broverman MIC 2.7(b)     | \$12k / 15% / 4yr                 | 30/360 | RE  | HU  | M              | M    | —      |
| CFPB H-25(B)             | \$162k / 3.875% / 30yr            | 30/360 | RE  | HU  | M              | M    | —      |
| CFPB IO Sample           | \$211k / 4% / 30yr                | 30/360 | RE  | HU  | M              | M    | —      |
| eCampus §4.3 Pearlline   | \$10k / 10% / 4yr                 | 30/360 | RE  | HU  | A              | A    | —      |
| eCampus §4.3 Erika       | \$32.6k / 4.83% / 9yr             | 30/360 | RE  | UP  | M              | M    | —      |
| eCampus §4.3 Johnetta    | \$20.2k / 3.53% / 8yr             | 30/360 | RE  | HU  | M              | M    | —      |
| eCampus §4.4.1 T1        | \$297.5k / 3.8% / 3yr, 20yr amort | 30/360 | RE  | HU  | j <sub>2</sub> | Q    | —      |
| eCampus §4.4.1 renew     | \$265.8k / 2.5% / 17yr            | 30/360 | RE  | HU  | j <sub>2</sub> | Q    | —      |
| FNMA §1103               | \$25M / 5.5% / 10yr, 30yr amort   | A/360  | RE  | HU  | M              | M    | —      |
| FHLBB 1935 Plan A        | \$3k / 6% / 139mo                 | 30/360 | CP  | HU  | M              | M    | —      |
| Geltner Ch 20            | \$1M / 12% / 30yr                 | 30/360 | CP  | HU  | M              | M    | —      |
| Goldstein §10.3          | \$563 / 12% / 5mo                 | 30/360 | CP  | HU  | M              | M    | —      |
| JHF Flat 35              | \$20M / 1.5% / 30yr               | 30/360 | RE  | HU  | M              | M    | —      |
| Las Positas §8.05 Ex 1   | \$15k / 9% / 5yr                  | 30/360 | RE  | UP  | M              | M    | —      |
| Las Positas §8.05 Ex 3   | \$18k / 2% / 5yr                  | 30/360 | RE  | UP  | M              | M    | —      |
| LoanKeisan JP            | \$30M / 1% / 30yr                 | 30/360 | RE  | DN  | M              | M    | —      |
| MoneyVox FR              | \$10k / 5% / 1yr                  | 30/360 | RE  | HU  | M              | M    | —      |
| MS State P3920           | \$100k / 7% / 30yr                | 30/360 | RE  | HU  | M              | M    | —      |
| Olivier Chans T1         | \$350.1k / 4.9% / 3yr, 20yr amort | 30/360 | RE  | HU  | j <sub>2</sub> | M    | —      |
| Olivier Chans renew      | \$316.6k / 5.85% / 17yr           | 30/360 | RE  | HU  | j <sub>2</sub> | M    | —      |
| OpenStax §6.12.110       | \$132.7k / 4.8% / 30yr            | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.100.1      | \$18.3k / 6.75% / 4yr             | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.100.2      | \$41.6k / 3.9% / 6yr              | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.110        | \$153.9k / 4.21% / 20yr           | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.114        | \$159.2k / 5.75% / 30yr           | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.36         | \$23.7k / 4.76% / 5yr             | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.78.1       | \$17.9k / 7.5% / 10yr             | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax AK 6.78.2       | \$33.8k / 4.3% / 20yr             | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax §6.8 car        | \$28.5k / 3.99% / 5yr             | 30/360 | RE  | UP  | M              | M    | —      |
| OpenStax §6.8 home       | \$136.7k / 5.75% / 15yr           | 30/360 | RE  | UP  | M              | M    | —      |
| ProEducate cap           | \$65k / 10% / 30yr                | 30/360 | RE  | HU  | M              | M    | 1, cap |
| RBC Acc Bi-Weekly        | \$350k / 5% / 25yr                | 30/360 | RE  | HU  | j <sub>2</sub> | BW   | —      |
| Reg Z H-14               | \$10k / 17.41% / 30yr             | 30/360 | RE  | HU  | M              | M    | 14     |
| Skinner §42 Ex 1         | \$1k / 6% / 15yr                  | 30/360 | RE  | HU  | A              | A    | —      |
| Skinner §42 Ex 3 piano   | \$500 / 6% / 5yr                  | 30/360 | RE  | HU  | A              | M    | —      |
| Solution Bank IT         | \$200k / 7% / 30yr                | 30/360 | RE  | HU  | M              | M    | —      |
| Synthetic HALF EVEN      | \$100k / 4.80% / 30yr             | 30/360 | RE  | HU  | M              | M    | —      |
| Wynn and Mingus Palc     | \$100k / 4.50% / 30yr             | 30/360 | RE  | HU  | M              | M    | —      |

45 fixtures total.

**Column codes.** Day: 30/360 US residential, A/360 Actual/360 US commercial. Bal: RE round-each-balance, CP carry-precision. Rnd: UP ROUND\_UP, DN ROUND\_DOWN, HU ROUND\_HALF\_UP, HE ROUND\_HALF\_EVEN. Cmp: M monthly, j semi-annual (Canadian *Interest Act* §6), A annual. Freq: M SM BW W Q A (monthly to annual). ARM: count of RateChange entries; cap indicates at least one payment\_cap\_factor. A dash is fully fixed-rate.

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**BCC Brescia Italy.** BCC di Brescia foglio informativo ‘Mutuo Fondiario Agrario Tasso Variabile Euribor’, Tipo Piano Francese, rata mensile for €200,000 at 5.65% over 10 years. <https://www.bccbrescia.it/umbraco/surface/transparency/ShowFile/67119>

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**LoanKeisan Japan.** LoanKeisan.com. “3,000 国庫庫 1.000 国庫庫 30 国庫庫.” Complete 360-row monthly amortization schedule. <https://www.loankeisan.com/bin/calc?a=3000&b=0&i=1&p=30&t=r>

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**Synthetic HALF\_UP / ROUND\_UP / HALF\_EVEN.** Synthetic boundary case constructed for this library: \$100,001.25 / 4.8% / 30yr is engineered so month-1 unrounded interest equals exactly \$400.005 — a half-cent boundary that distinguishes ROUND\_HALF\_UP (\$400.01) from ROUND\_HALF\_EVEN (\$400.00) and from ROUND\_UP (which rounds the payment up to \$524.68 instead of \$524.67).

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**Zero-Interest Promotional Financing.** Synthetic fixture modeling 0% APR promotional financing (e.g. consumer electronics or furniture loans).